

Transient Analysis: Uncompensated VCO Calibration and Corners Flow

1. Testbench Configuration

Testbench: PrimeSim_default.VCO_uncomp

Design: Tese/TransientAnalysisVCO/schematic_uncomp

Analysis: Transient (tran) from 0 to 200 ns with a 20 ps step.

2. Output Expressions

OUTPUT	EXPRESSION
V_{drive}	v(/I136/vdrive)
$V_{co,p}$ (vco_p)	v(/vco_p)
Average Frequency	avgfreq(vco_p, 100n, 200n)
I_{tail}	i(/I136/i_tail)
$V_{ref,filt}$	v(/I136/magamp/vref_filt)
Coarse Tune	v(/coarse_tune<6:0>)
$V_{reg,vco}$	v(/I136/magamp/vreg_vco)
V_{amp}	vreg_vco - vref_filt
I_{tail} (Internal I74)	i(/I136/I74/i_tail)
$I_{tail,smooth}$	smooth(o(I_tail, analysisName="tran"), 1n)

3. Calibration & Corner Setup

A calibration phase is performed across three core process corners (Typical, Best-Case Q, and Worst-Case Q), sweeping the control voltage V_{ref} to establish the optimal discrete operating points. The resulting calibrated V_{ref} vectors are then mapped to extreme temperature conditions (-40°C and 125°C) to form the complete set of transient corners.

CORNER	TEMP	CALIBRATED V_{ref} SWEEP
Typm40	-40°C	{19m 36m 52m 66m 79m 90m 100m 109m 116m 122m}
Typ125	125°C	{19m 36m 52m 66m 79m 90m 100m 109m 116m 122m}
bcQm40	-40°C	{23m 41m 56m 71m 84m 95m 105m 113m 120m 125m}
bcQ125	125°C	{23m 41m 56m 71m 84m 95m 105m 113m 120m 125m}

CORNER	TEMP	CALIBRATED V_{ref} SWEEP
wcQm40	-40°C	{16m 33m 48m 62m 74m 86m 96m 104m 112m 118m}
wcQm125	125°C	{16m 33m 48m 62m 74m 86m 96m 104m 112m 118m}